

System Overview

Music Podcast Player

1 INTRODUCTION

The Music Podcast Player is a discoverability-focused, web-based audio player built around RSS feeds. Artists who host their own RSS feeds can distribute music independently, without relying on centralized platforms like Spotify or Apple Music, allowing them to retain direct control over their content. Music Podcast Player fetches those feeds, then presents them in a clean, accessible way. This document provides a high-level overview of the system for readers who may be unfamiliar with the project.

2 SYSTEM DESCRIPTION

Music Podcast Player lets users discover and listen to music distributed via RSS feeds. Users can search for feeds by keyword or input a feed URL directly. Once a feed is selected, its full track listing is displayed, alongside album art and artist information. Selecting any track begins automatic playback.

Authenticated users can organize their listening with playlists. Tracks can be added to existing playlists or used to create new ones. Playlists are tied to user profiles, which can be set to public or private and shared with others.

Music Podcast Player also supports direct artist payments through the Lightning Network. While a track is playing, users can send a one-time Boost, optionally including a short message, or enable Sat Dripping to automatically send small recurring payments at set intervals while they listen.

3 KEY FEATURES

- **Desktop and Mobile Support** – Music Podcast Player can be accessed via both desktop and mobile platforms.
- **Keyword Searching** – On the "Search" page, users can search the Podcast Index to discover feeds.
- **Direct Feed Input** – On the "Input" page, users can paste in a specific feed's URL to pull up that feed.
- **Boostagrams** – Users can send an artist a one-time payment, called a boost, to support an artist directly.
- **Sat Dripping** – Users can enable Sat Dripping, which sends smaller payments (in satoshis, a Bitcoin unit) to artists at regular intervals while listening.

- **User Profiles and Playlists** – Users can create their own profile, where they can create, share, and listen to custom playlists.
- **Modern Playback Controls** – Music Podcast Player includes standard playback controls: Play/Pause, Step Forward/Backward, Repeat Feed, Reverse Feed Order, Shuffle, and Restart Song.

4 ARCHITECTURE OVERVIEW

Figure 1 shows the high-level architecture of Music Podcast Player. The frontend, built in Vue 3, communicates with a Flask/FastAPI backend, which handles feed fetching from the Podcast Index API, as well as user and playlist data stored in PostgreSQL. The entire system is hosted on AWS, with GitHub Actions managing automated deployment.

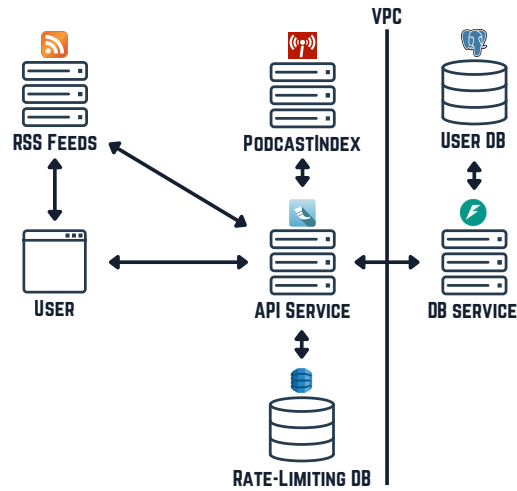


Figure 1: Music Podcast Player’s Architecture

5 SYSTEM BOUNDARIES AND INTEGRATIONS

Music Podcast Player is responsible for feed discovery, media playback, user accounts, and playlist management. It does not host or store any audio content. Artists host their own feeds externally, and the app fetches and parses them on-demand via the Podcast Index API.

The system works with three external services:

- **Podcast Index API** – Feed search and discovery. Music Podcast Player queries this API to retrieve feed metadata and tracks.
- **Lightning Network** – Handles all payment processing for Boosts and Sat Dripping. Wallet authentication and transaction signing are managed externally.

- **Amazon Web Services (AWS)** – Hosts the application and its infrastructure.

User authentication and playlist data are managed internally and stored in the application’s PostgreSQL database.

6 TECHNOLOGY STACK

Music Podcast Player is built on a modern web stack across frontend, backend, data, and infrastructure layers. See Table 1.

Category	Technology
Frontend	Vue 3
Backend	Flask, FastAPI
Database	PostgreSQL
Cloud Infrastructure	Amazon Web Services (AWS)
CI/CD	GitHub Actions

Table 1: Tech Stack

7 GLOSSARY

Boost / Boostagram A one-time, user-defined payment sent directly to an artist via the Lightning Network.

CI/CD Continuous Integration / Continuous Deployment. A DevOps practice that automates building, testing, and deploying software.

Feed URL The direct URL pointing to an RSS feed file, used to load a specific feed into Music Podcast Player.

Lightning Network A payment protocol built on top of the Bitcoin blockchain, designed for fast micropayments.

Podcast Index An open, third-party database of podcast and music RSS feeds. Music Podcast Player uses its API for feed discovery and search.

RSS Feed (Really Simple Syndication) A standardized file format that allows content creators to publish and distribute media, like music or podcasts, independently, without a centralized platform.

Sat Dripping A passive support feature that automatically sends small, recurring Bitcoin micropayments to an artist at set intervals during playback.

Satoshi (Sat) The smallest unit of Bitcoin (100 million satoshis = 1 BTC), used for micropayments over the Lightning Network.